



2.3 Estimating Derivatives of a Function at a Point

2.4 Connecting Differentiability and Continuity: Determining When Derivatives Do and Do Not Exist

Ex 1: What is the average rate of change of $y = \cos(2x)$ on the interval $\left[0, \frac{\pi}{2}\right]$?

(A) $-\frac{4}{\pi}$

(B) -1

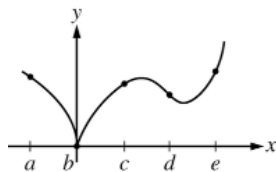
(C) 0

(D) $\frac{\sqrt{2}}{2}$

(E) $\frac{4}{\pi}$

$$\frac{\cos\left(2 \cdot \frac{\pi}{2}\right) - \cos(2 \cdot 0)}{\frac{\pi}{2} - 0} = \frac{-1 - 1}{\frac{\pi}{2}} = -\frac{4}{\pi}$$

Ex 2: The graph of the function f is shown in the figure. For which of the following values of x is $f'(x)$ positive and increasing?



Graph of f

(A) a

(B) b

(C) c

(D) d

(E) e

Ex 3: Use the alternate form to find the slope at the indicated point, if possible.

a) $g(x) = \sqrt[3]{x}$, $(0, 0)$

$$g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0}$$

$$g'(0) = \lim_{x \rightarrow 0} \frac{\sqrt[3]{x} - 0}{x - 0}$$

$$g'(0) = \lim_{x \rightarrow 0} \frac{1}{x^{\frac{2}{3}}} = \infty$$

$$g'(0) \text{ DNE}$$

b) $h(x) = \sqrt{x}$, $(0, 0)$

$$h'(0) = \lim_{x \rightarrow 0} \frac{h(x) - h(0)}{x - 0}$$

$$h'(0) = \lim_{x \rightarrow 0} \frac{\sqrt{x} - 0}{x - 0}$$

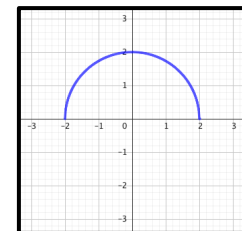
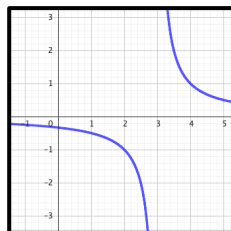
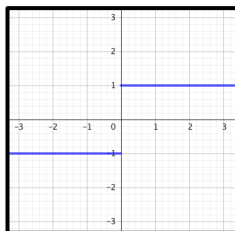
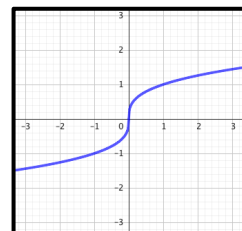
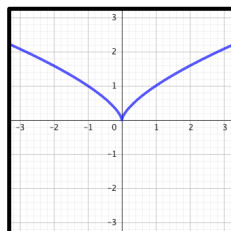
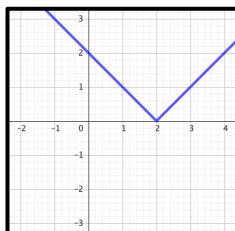
$$h'(0)^- = \lim_{x \rightarrow 0^-} \frac{1}{x^{\frac{1}{2}}} \text{ DNE}$$

$$h'(0) \text{ DNE}$$

Differentiability and Continuity

Functions are *not* differentiable at

1. sharp turns (cusps/corners)
2. vertical tangents
3. discontinuities
4. endpoints



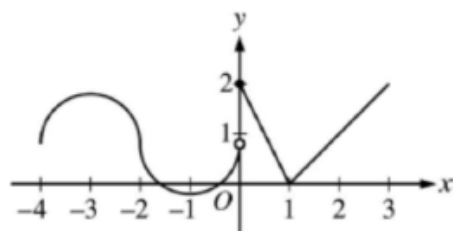
A consequence of differentiability is continuity. That is, if f is differentiable at $x = c$, f is also continuous at $x = c$.



A continuous function may NOT be differentiable!



Ex 4:



Graph of f

- (A) $x = 1$
(B) $x = -2$ and $x = 0$
(C) $x = -2$ and $x = 1$
(D) $x = 0$ and $x = 1$

The graph of the piecewise-defined function f is shown in the figure above. The graph has a vertical tangent line at $x = -2$ and horizontal tangent lines at $x = -3$ and $x = -1$. What are all values of x , $-4 < x < 3$, at which f is continuous but not differentiable?

Ex 5: If g is a function for which $\lim_{h \rightarrow 0} \frac{g(3+h) - g(3)}{h} = 0$, which of the following statements must be true?

- | | | |
|----------------|----------------------------------|------------------------------------|
| I. $g'(3) = 0$ | II. g is continuous at $x = 3$ | III. g' is continuous at $x = 3$ |
|----------------|----------------------------------|------------------------------------|

- (A) I only
(B) II only
(C) I and II only
(D) II and III only
(E) I, II and III

Approximating Derivatives with Tables

To approximate a derivative of f at $x = c$ using tabular data, find the smallest interval containing $x = c$, then find the average rate of change on that interval.

Ex 6: The table shows $f(x)$ for several values of x , where f is a differentiable function of x . Estimate the instantaneous rate of change of f at $x = 5$.

x	0	2	4	6
$f(x)$	0	10	35	75

$$f'(5) \approx \frac{f(6) - f(4)}{6 - 4} = \frac{75 - 35}{6 - 4} = \frac{40}{2} = 20$$

*answers do NOT need to be simplified

Ex 7: 2024 AB 1 part (a)

t (minutes)	0	3	7	12
$C(t)$ (degrees Celsius)	100	85	69	55

$$C'(5) \approx \frac{C(7) - C(3)}{7 - 3} = \frac{69 - 85}{7 - 3} \text{ degrees Celsius/minute}$$

*answers do NOT need to be simplified

The temperature of coffee in a cup at time t minutes is modeled by a decreasing differentiable function C , where $C(t)$ is measured in degrees Celsius. For $0 \leq t \leq 12$, selected values of $C(t)$ are given in the table shown.

- (a) Approximate $C'(5)$ using the average rate of change of C over the interval $3 \leq t \leq 7$. Show the work that leads to your answer and include units of measure.



Approximating Derivatives Using Technology

Ex 8: Use a graphing utility to find the indicated derivative.

a) $f(x) = \ln(x^2 - 4x + 5)$, $f'(1.1)$

NORMAL FLOAT AUTO REAL RADIAN MP

Plot1 Plot2 Plot3
Y1=ln(x^2-4x+5)
Y2=
Y3=
Y4=
Y5=
Y6=
Y7=
Y8=

NORMAL FLOAT AUTO REAL RADIAN MP

d/dx(Y1)|x=1.1
-0.9944749165

$$f'(1.1) \approx 0.994$$

b) $y = 7.6 \tan^{-1}(0.2x)$, $\frac{dy}{dx} \Big|_{x=6}$

NORMAL FLOAT AUTO REAL RADIAN MP

d/dx(7.6tan^-1(.2X))|x=6
0.6229508243

$$\frac{dy}{dx} \Big|_{x=6} \approx 0.622 \text{ or } 0.623$$

*students can round OR truncate to three places after the decimal

Ex 9: The rate of flow of gasoline, in gallons per second, can also be modeled by

$g(t) = \left(\frac{t}{450}\right) \cos\left(\left(\frac{t}{100}\right)^2\right)$ for $0 \leq t \leq 150$. Using this model, find the value of $g'(140)$. Indicate units of measure.

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d/dx(((x/450)cos((x/100)^2))|x=140
-0.0089028465

$$g'(140) \approx -0.008 \text{ or } -0.009$$

Writing Equations of Tangent Lines

To write an equation of a tangent line, you need a point and a slope.

Ex 10: Write an equation of the tangent line to $f(x) = \ln(x^2 - 4x + 5)$ at $x = 1.1$.

NORMAL FLOAT AUTO REAL RADIAN MP

Plot1 Plot2 Plot3
Y1=ln(x^2-4x+5)
Y2=
Y3=
Y4=
Y5=
Y6=
Y7=
Y8=

NORMAL FLOAT AUTO REAL RADIAN MP

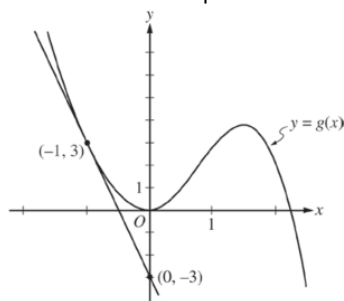
Y1(1.1)
0.5933268453

NORMAL FLOAT AUTO REAL RADIAN MP

d/dx(Y1)|x=1.1
-0.9944749165

$$y - 0.593 = -0.994(x - 1.1)$$

Ex 11: The figure shows the graph of the function g and the line tangent to the graph of g at $x = -1$. Write an equation of the tangent line to g at $x = -1$.



$$g(-1) = 3$$

$$g'(-1) = \frac{3 - -3}{-1 - 0} = -6$$

$$y - 3 = -6(x + 1)$$